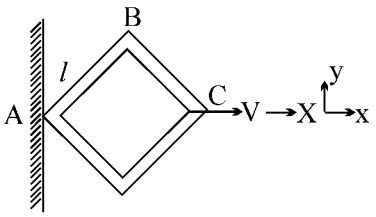
## SECTION-I

### Single Correct Answer Type

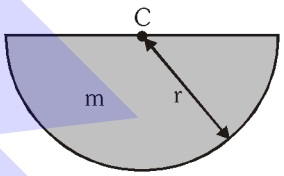
10 Q. [3 M (–1)]

**1.** Four rods side length $l$ have been hinged to form a rhombus. Vertex A is fixed to rigid support, vertex C is being moved along the X-axis with a constant velocity V as shown in figure. The rate at which vertex B is nearing the X-axis at the moment the rhombus is in the form of a square is



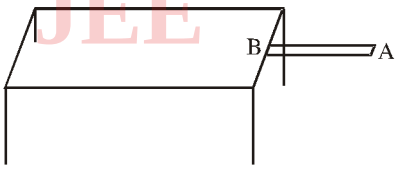

(A) V/4 (B) $\frac{V}{\sqrt{2}}$ (C) V/2 (D) $V^2/g$

**2.** A semicircular lamina of mass m has radius r and centre C. Its moment of inertia about an axis through its centre of mass and perpendicular to its plane is–



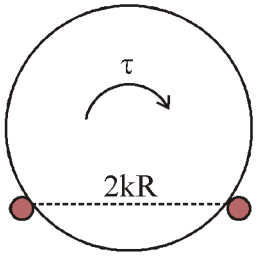

(A) $\frac{1}{2}mr^2$ (B) $\left(\frac{1}{2}-\frac{4}{\pi^2}\right)mr^2$ (C) $\left(\frac{1}{2}-\frac{8}{9\pi^2}\right)mr^2$ (D) $\left(\frac{1}{2}-\frac{16}{9\pi^2}\right)mr^2$

**3.** A uniform thin stick of length $\ell$ and mass m is held horizontally with its end B hinged at a point B on the edge of a table. Point A is suddenly released. The acceleration of the centre of mass of the stick at the time of release, is :-




(A) $\frac{3}{4}$ g (B) $\frac{3}{7}$ g (C) $\frac{2}{7}$ g (D) $\frac{1}{7}$ g

**4.** A cylinder of diameter 2R and mass M rests on two rough pegs (coefficient of static friction $\mu$), the distance 2kR apart. A gradually increasing torque $\tau$ is applied as shown. The cylinder will topple before it slips if




(A) $\mu < \frac{k}{\sqrt{1-k^2}}$ (B) $\mu > \frac{k}{\sqrt{1-k^2}}$

(C) $\mu > \sqrt{1-k^2}$ (D) $\mu < \sqrt{1-k^2}$

**5.** We want to build a clock having different features. Mark **INCORRECT** statement :-

(A) It is possible to build a clock in which the tips of the second hand, the minute hand, and the hour hand move with the same tangential acceleration.

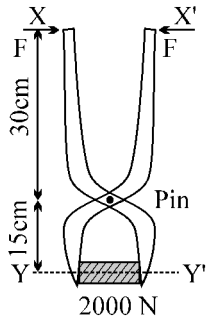
(B) It is possible to build a clock in which the tips of the second hand, the minute hand, and the hour hand move with the same angular acceleration.

(C) It is possible to have a situation in which the second hand, the minute hand, and the hour hand with the same angular position.

(D) It is possible to build a clock in which the tips of the second hand, the minute hand, and the hour hand move with the same angular speed.


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**6.** Figure shows a pair of pin jointed gripper tongs holding an object weighing 2000 N. The coefficient of friction μ at the gripping surface is 0.1. X-X' is the line of action of the input force and Y-Y' is the line of application of normal gripping force. If the pin-joint is assumed as frictionless, the magnitude of force F required to hold the weight is




(A) 1000 N

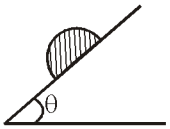
(B) 2000 N

(C) 2500 N

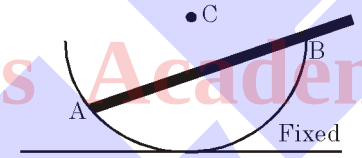
(D) 5000 N

**7.** An uniform hemi-solid sphere is placed with flat surface on rough inclined plane as shown in figure. If friction is large for no sliding, then the minimum angle θ at which toppling occur is

(A) $\tan^{-1}\left(\frac{1}{2}\right)$ (B) 45° (C) $\tan^{-1}\left(\frac{8}{3}\right)$ (D) $\tan^{-1}\left(\frac{4}{3}\right)$




**8.** A uniform rod is placed inside a smooth hemispherical fixed bowl as shown and released. Then which of the option is **CORRECT** :-




(A) The line of action of normal reaction at B, does not pass through centre (C).

(B) The rod can never stay in equilibrium.

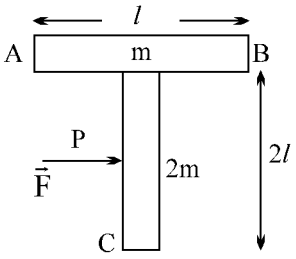
(C) In the free body diagram of the rod, none of the forces on the rod is parallel or perpendicular to the rod.

(D) None of these

**9.** A ring of radius R is rolling purely on the outer surface of a pipe of radius 4R. At some instant, the center of the ring has a constant speed = v. Then , the acceleration of the point on the ring which is in contact with the surface of the pipe is :

(A) $4v^2/5R$ (B) $3v^2/5R$ (C) $v^2/4R$ (D) zero

**10.** A T shaped object with dimensions shown in the figure is lying on a smooth floor. A force $\vec{F}$ is applied at the point P parallel to AB such that the object has only the translational motion without rotation. Find the location of P with respect to C.




(A) $(3/4)l$

(B) $l$

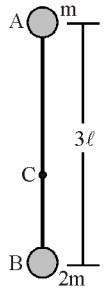
(C) $(4/3)l$

(D) $(3/2)l$


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## Multiple Correct Answer Type 5 Q. [4 M (–1)]

**11.** A light rigid rod AB of length $3\ell$ has a point mass m at end A and a point mass 2m at end B is kept on a smooth horizontal surface. Point C is the center of mass of the system. Initially the system is at rest. The mass 2m is suddenly given a velocity $v_0$ towards right. Take Z axis to be perpendicular to the plane of the paper.



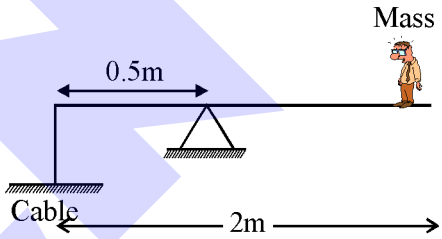

(A) The minimum moment of inertia (about Z-axis), $I_{zz}$ of the system is $5m\ell^2$.

(B) The magnitude of tension in the rod in subsequent motion is $\frac{2mv_0^2}{9\ell}$

(C) The ratio of moment of inertia about Z-axis at points A and B, $\frac{I_{zz}^A}{I_{zz}^B} = 2$

(D) Point C remains stationary during subsequent motion.

**12.** A man of mass 85 kg stands on one end of a uniform board of length 2 m and mass 15 kg. The other end is tied to the floor by a steel cable, and the board is supported from below just 0.5 m from the cable end (see figure).



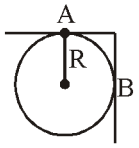

(A) The tension in cable is 1000 N

(B) The tension in cable is 2700 N

(C) The normal reaction at support is 3700 N

(D) The normal reaction at support is 1700 N

**13.** A rod bent at right angle along its centre line, is placed on a rough horizontal fixed cylinder of radius R as shown in figure. Mass of rod is 2 m and rod is in equilibrium. Assume that friction force on rod at A and B are equal in magnitde.



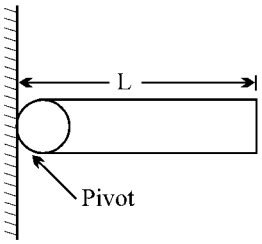

(A) Normal force applied by cylinder on rod at A is 3mg/2.

(B) Normal force applied by cylinder on rod at B must be zero.

(C) Friction force acting on rod at B is upward.

(D) Normal force applied by cylinder on rod at A is mg.

**14.** A uniform rod of length L and mass M is free to rotate about a frictionless pivot at one end as shown in the figure. The rod is released form rest in the horizontal position



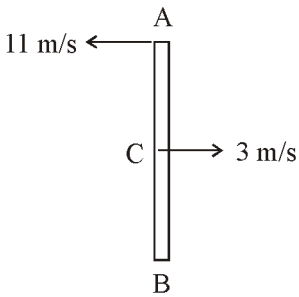

(A) initial angular acceleration of right end of the rod is 3g/2L and is common to all points on the rod.

(B) initial linear acceleration of the right end of the rod is 3g/2 and is common to all points on the rod.

(C) initial linear acceleration of the right end of the rod is 3g/2 and its value is different at different points.

(D) no points on the rod have a linear acceleration greater than g.

**15.** A uniform rod AB of length 7m is undergoing combined rotational and translational motion on smooth horizontal surface such that, at some instant of time, velocities of its end point A and centre C are both perpendicular to the rod and opposite in direction, having magnitude 11 m/s and 3 m/s respectively as shown in the fig. Now consider that no external force and torque is acting on the rod. Then select **CORRECT** options.




(A) acceleration of point A is 56 m/s$^2$

(B) acceleration of point B is 56 m/s$^2$

(C) at the instant shown in the figure acceleration of point C is zero.

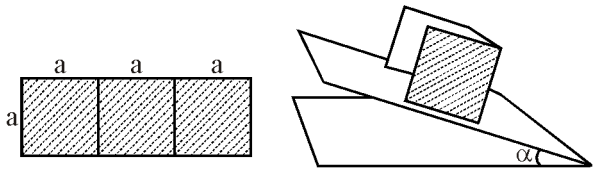
(D) angular velocity of the rod is 4 rad/s


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**Linked Comprehension Type (1 Para × 3Q.) (1 Para × 2Q.) [3 M (-1)]**
**(Single Correct Answer Type)**

**Paragraph for Question nos. 16 to 18**

A cardboard strip, bent in the shape of the letter C, is put on a rough inclined plane, as shown in the figure.




**16.** At what angle of inclination to the horizontal plane will it topple? (assume that it does not slide).

(A) $\tan^{-1}\left(\frac{2}{3}\right)$ (B) $\tan^{-1}\left(\frac{3}{2}\right)$ (C) 45° (D) $\tan^{-1}\left(\frac{1}{3}\right)$

**17.** What should be the coefficient of friction so that it does not slide before toppling.

(A) 0.66 (B) 0.75 (C) 1.0 (D) 0.33

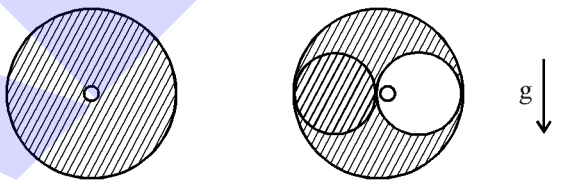
**18.** What should be the time taken to travel a distance a if the inclined plane is made completely smooth? [Take angle of inclined plane as $\alpha$]

(A) $\sqrt{\frac{a}{g\sin\alpha}}$ (B) $\sqrt{\frac{2a}{g\sin\alpha}}$ (C) $\sqrt{\frac{2a}{3g\sin\alpha}}$ (D) None of these

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**Paragraph for Question no. 19 and 20**

A disk of radius 2R and mass 'M' is hinged from its center. If a disk of radius R been removed such that radius of bigger disk behave as a diameter of the disk and it is welded on the opposite side as shown in diagram.




Case-I Case-II

**19.** Find the ratio of moment of inertia about point 'O' in Case-I and Case-II :-

(A) $\frac{1}{2}$ (B) $\frac{2}{1}$ (C) $\frac{1}{\sqrt{2}}$ (D) 1

**20.** Find the angular acceleration of the system in case-II if the system is released from the given position about the axis passing through O and perpendicular to the plane of disc:-

(A) $\frac{g}{R}$ (B) $\frac{g}{2R}$ (C) $\frac{g}{3R}$ (D) $\frac{g}{4R}$

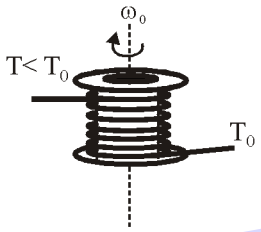

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## SECTION-II

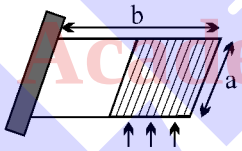
**Numerical Answer Type Question** **2Q.[3M(0)]**

**(upto second decimal place)**

**21.** A capstan is a rotating drum (cylinder) over which a rope or cord slides in order to increase the tension due to friction. If the difference in tension between the two ends of the rope is 500 N and the capstan has a diameter of 10 cm and rotates with angular velocity 10 rad/s. Capstan is made of iron and has mass 5 kg, specific heat 1000 J/kg K. At what rate does temperature rise ? Assume that the temperature in the capstan is uniform and all the thermal energy generated flows into it. Express your answer as $x \times 10^{-4}$ °C/s. Fill up value of x.




**22.** There is a rectangular plate of mass M kg of dimensions (a × b). The plate is held in horizontal position by striking n small balls each of mass m per unit area per unit time. These are striking in the shaded half region of the plate. The balls are colliding elastically with velocity v. What is v? It is given n = 100, M = 3 kg, m = 0.01 kg; b = 2 m; a = 1m; g = 10 m/s$^2$. **[JEE 2006]**

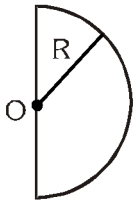



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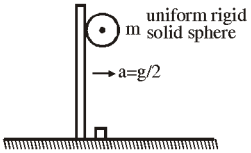
## SECTION-III

**Numerical Grid Type (Ranging from 0 to 9)** **2 Q. [4 M (0)]**

**23.** A uniform semicircular disc of mass 'M' and radius 'R' hinged at point O a shown in figure is released from rest from a vertical position as shown. If the initial magnitude of angular acceleration of the disc is $\frac{xg}{3\pi R}$ than the value of x is (Center of mass is at a distance $\frac{4R}{3\pi}$ from center of disc):




**24.** A uniform solid sphere rolls down on a vertical rigid surface without slipping. The vertical surface moves with an acceleration a = g/2 in horizontal direction as shown. The minimum coefficient of friction between the sphere and vertical surface so as to prevent slipping is μ. Write 14 μ as answer.



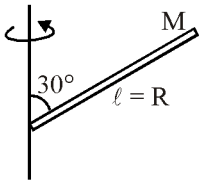
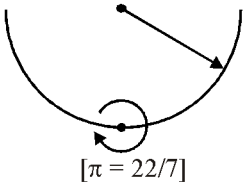
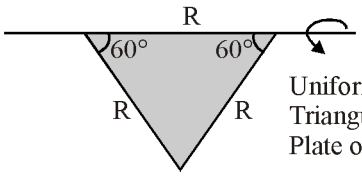
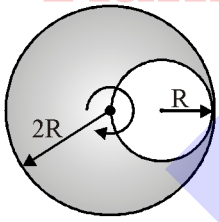


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## SECTION-IV

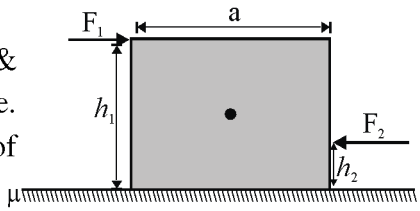
**Matrix Match Type (4 × 5)** **3 Q. [8 M (for each entry +2(0)]**

**25.**

| | Column-I | | Column-II | |
|---|---|---|---|---|
| (A) |   Uniform Rod | (P) | $\frac{8MR^2}{11}$ |
| (B) |   Uniform Semicircular Ring. Axis is perpendicular to plane of ring. | (Q) | $\frac{MR^2}{12}$ |
| (C) |   Uniform Triangular Plate of mass M | (R) | $\frac{13MR^2}{8}$ |
| (D) |   Uniform disk of initial mass M from which circular portion of radius R is then removed. M.I. of remaining mass about axis which is perpendicular to plane of plate | (S) | $\frac{MR^2}{8}$ |

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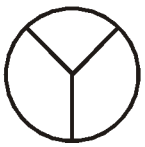
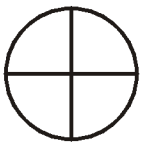
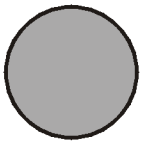
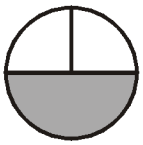
**26.** Consider a square plate of mass 'm' kept on a rough horizontal ground with friction coefficient μ=1. Two anti parallel forces $F_1$ & $F_2$ is applied at height $h_1$ & $h_2$ respectively as shown in figure. Column I gives various values of forces & height of point of application of that force while column II gives various effects.




| Column I | Column II |
|---|---|
| (A) $F_1=F_2=\frac{mg}{2}$; $h_1=a$; $h_2=\frac{a}{8}$ | (P) Normal reaction shifts towards right |
| (B) $F_1=\frac{mg}{2}$; $F_2=mg$; $h_1=a$; $h_2=\frac{a}{8}$ | (Q) Torque due to friction is into the plane of plate |
| (C) $F_1=mg$; $F_2=0$; $h_1=a$ | (R) Torque due to friction is out of the plane of plate |
| (D) $F_1=\frac{mg}{2}$; $F_2=mg$; $h_1=h_2=\frac{a}{2}$ | (S) Toppling of plate occurs |
| | (T) Friction force is zero. |


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**27.** Each of the four wheels in column I has an outer ring having radius R and mass m. Other than the outer ring the wheels comprise of some uniform rods (each of mass m and length R) or some lamina (having the same mass m). Column II gives radius of gyration about axis passing through centre of ring perpendicular to its plane.

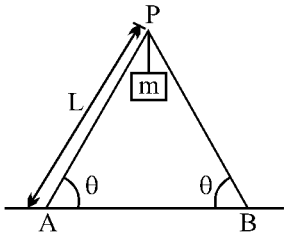
| **Column–I : Wheels** | | **Column–II : Radius of gyration** | |
|---|---|---|---|
| (A) |  | (P) | $R\sqrt{\frac{3}{4}}$ |
| (B) |  | (Q) | $R\sqrt{\frac{11}{18}}$ |
| (C) |  One disc + ring | (R) | $\frac{R}{\sqrt{2}}$ |
| (D) |  Outer ring + half disc (having mass m) + one rod | (S) | $R\sqrt{\frac{7}{15}}$ |
| | | (T) | $\frac{R}{\sqrt{3}}$ |

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## Subjective Type

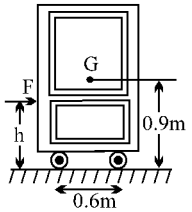
**3 Q. [4 M (0)]**

**28.** Two identical ladders, each of mass M and length L are resting on the rough horizontal surface as shown in the figure. A block of mass m hangs from P. If the system is in equilibrium, find the magnitude and the direction of frictional force at A and B.




**29.** Find the MI of a rod about an axis through its centre of mass and perpendicular to the length whose linear density varies as $\lambda = ax$ where a is a constant and x is the position of an element of the rod relative to its left end. The length of the rod is $\ell$.

**30.** A 20 kg cabinet is mounted on small casters that allow it to move freely ($\mu = 0$) on the floor. If a 100 N force is applied as shown, determine




(a) the acceleration of the cabinet,

(b) the range of values of h for which the cabinet will not tip.


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GR # ROTATIONAL DYNAMICS-1 ANSWER KEY

## SECTION-I

**Single Correct Answer Type** **10 Q. [3 M (–1)]**

1. Ans. (C) 2. Ans. (D) 3. Ans. (A) 4. Ans. (B) 5. Ans. (D) 6. Ans. (D)
7. Ans. (C) 8. Ans. (A) 9. Ans. (A) 10. Ans. (C)

**Multiple Correct Answer Type** **5 Q. [4 M (–1)]**

11. Ans. (BC) 12. Ans. (B, C) 13. Ans. (A,C) 14. Ans. (A,C) 15. Ans. (A,B,C,D)

**Linked Comprehension Type** **(1 Para × 3Q.) (1 Para × 2Q.) [3 M (-1)]**

**(Single Correct Answer Type)**

16. Ans. (A) 17. Ans. (A) 18. Ans. (B) 19. Ans. (D) 20. Ans. (D)

## SECTION-II

**Numerical Answer Type Question** **2Q.[3M(0)]**

**(upto second decimal place)**

21. Ans. 500.00 22. Ans. 10 m/s

## SECTION-III

**Numerical Grid Type (Ranging from 0 to 9)** **2 Q. [4 M (0)]**

23. Ans. 8 24. Ans. 8

## SECTION-IV

**Matrix Match Type (4 × 5)** **3 Q. [8 M (for each entry +2(0)]**

25. Ans. (A) → (Q) ; (B) → (P) ; (C) → (S) ; (D) → (R)
26. Ans. (A) → (P,T) ; (B) → (P,R) ; (C) → (P,Q,S) ; (D) → (R)
27. Ans. (A) → (R) ; (B) → (S) ; (C) → (P) ; (D) → (Q)

**Subjective Type** **3 Q. [4 M (0)]**

28. Ans. $f = (M+m)\, g \dfrac{\cot\theta}{2}$ 29. Ans. $\dfrac{a\ell^4}{36}$ 30. Ans. (a) 5 m/s² →, (b) $0.3 < h < 1.5$ m


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